



Optimization and Algorithms in Sparse Regression

A Summary of My Thesis

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Overarching Theme

Practical concerns in sparse regression: speed, accuracy, and reproducibility.

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Overview

I–III Screening rules

IV Benchmarking optimization methods

V Coordinate descent (an optimization method) for SLOPE

VI Normalization and regularization

Screening Rules (Papers I–III)

- Data sets are growing in size.

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- Hyperparameter tuning increases costs further.

Basic Insight

- The support set is small in sparse regression, especially when $p \gg n$.¹
- Can we somehow figure this support **before** fitting the model?

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General Idea

1. Estimate how likely it is that a feature is in the support set.
2. If unlikely, discard it.
3. Fit a reduced model.
4. If we were wrong, just refit the model with missing features added.

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Turns out to be a pretty good idea!

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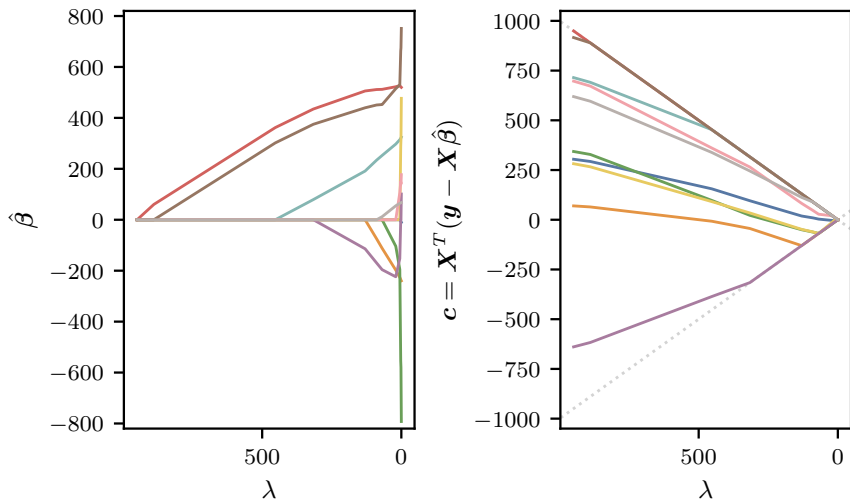


Figure 1: Coefficients and correlations (c) along the lasso path

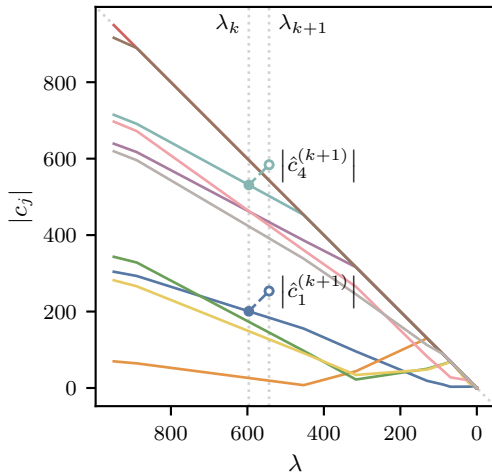


Figure 2: The strong rule for the lasso (Tibshirani et al. 2012). $|\hat{c}_j^{(k+1)}| = |c_j^{(k)}| + \lambda_k - \lambda_{k+1}$ is the absolute correlation estimate for feature j at step $k + 1$.

Strong Rule for SLOPE (Paper I)

Same idea as for lasso strong rule! Just need the subdifferential.

SLOPE Subdifferential

The set of all $\mathbf{g} \in \mathbb{R}^p$ such that

$$\mathbf{g}_{\mathcal{A}_i} = \left\{ \mathbf{s} \in \mathbb{R}^{|\mathcal{A}_i|} \mid \left\{ \begin{array}{ll} \text{cumsum}(|\mathbf{s}|_{\downarrow} - \boldsymbol{\lambda}_{R(\mathbf{s})_{\mathcal{A}_i}}) \preceq \mathbf{0} & \text{if } \beta_{\mathcal{A}_i} = \mathbf{0}, \\ \text{cumsum}(|\mathbf{s}|_{\downarrow} - \boldsymbol{\lambda}_{R(\mathbf{s})_{\mathcal{A}_i}}) \preceq \mathbf{0} \\ \text{and } \text{sign}(\beta_{\mathcal{A}_i}) = \text{sign}(\mathbf{s}) \\ \text{and } \sum_{j \in \mathcal{A}_i} (|s_j| - \lambda_{R(\mathbf{s})_j}) = 0 & \text{otherwise.} \end{array} \right. \right\}$$

- $\mathcal{A}_i$ defines a **cluster** containing indices of coefficients equal in absolute value.
- $R(\mathbf{x})$ returns **ranks** of absolute values of elements in $\mathbf{x}$.
- $|\mathbf{x}|_{\downarrow}$ returns the absolute values of $\mathbf{x}$ sorted in non-increasing order.

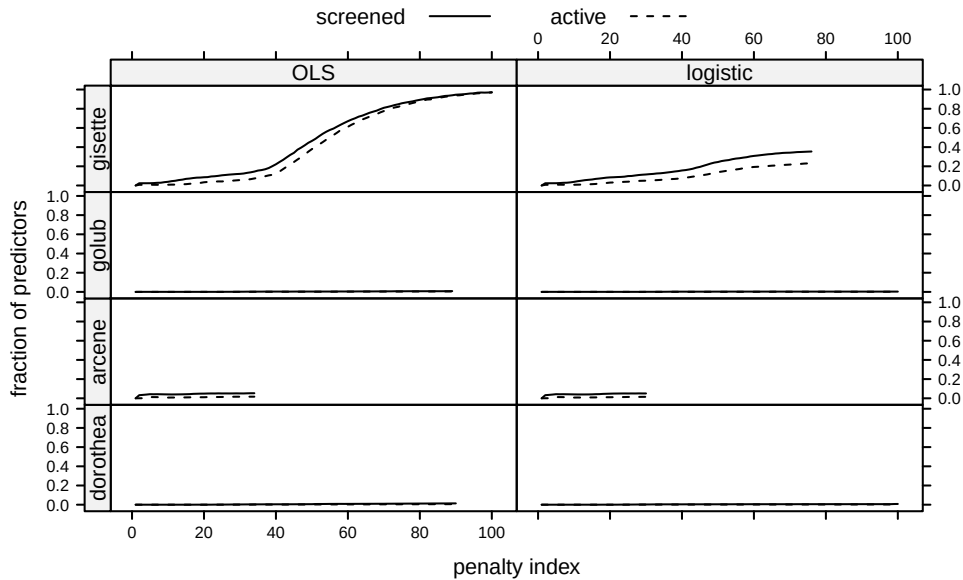


Figure 3: Effectiveness for some real data sets

Table 1: Time to fit lasso paths using either the strong screening rule or no rule.

Dataset	Model	n	p	Time (s)	
				No Screening	Screening
dorothea	Logistic	800	88 119	914	14
e2006-tfidf	Least squares	3308	150 358	43 353	4944
news20	Multinomial	1000	62 061	5485	517
physician	Poisson	4406	25	34	34

Look-Ahead Screening (Paper II)

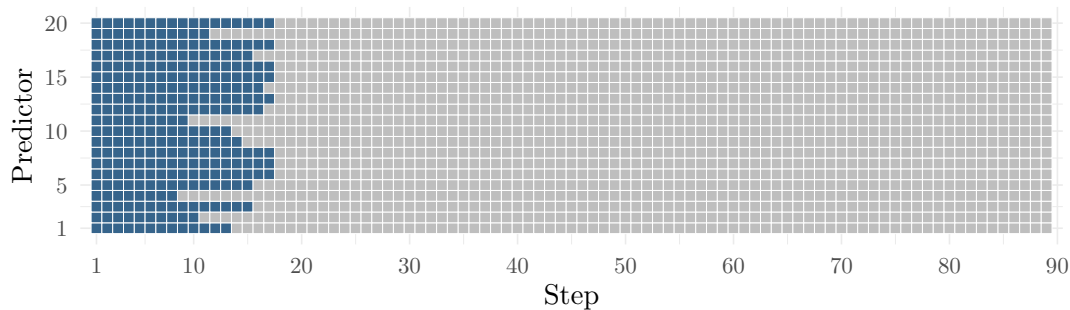


Figure 4: The lasso path for a sample of 20 features from the leukemia data set. The squares show for which steps the feature can be discarded.

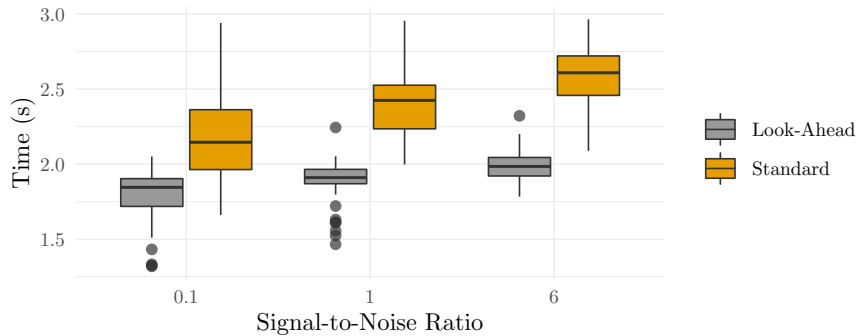


Figure 5: Benchmarks of time to fit the full lasso path with or without look-ahead screening.

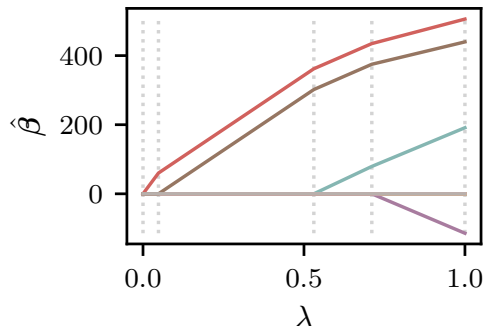
The Hessian Screening Rule

For the ordinary lasso,

$$g(\beta) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|_2^2,$$
$$\nabla g(\beta) = \mathbf{X}^\top (\mathbf{X}\beta - \mathbf{y}),$$

solution is **piecewise linear**.

And if active set is constant in $[\lambda_k, \lambda_{k+1}]$, we can express the solution in closed form.



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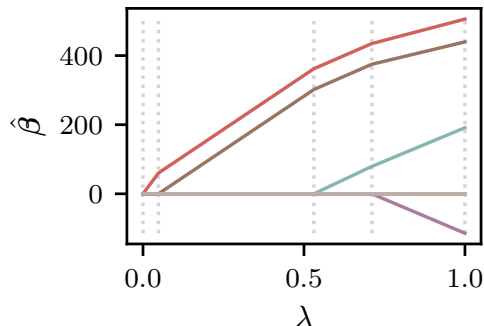
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Hessian Screening Rule (Basic Form)

Plug $\hat{\beta}$ into the gradient at step $k + 1$:

$$\tilde{\mathbf{c}}^H(\lambda_{k+1}) = -\nabla g(\hat{\beta}(\lambda_{k+1})_{\mathcal{A}}) = \mathbf{c}(\lambda_k) + (\lambda_{k+1} - \lambda_k) \mathbf{X}^\top \mathbf{X}_{\mathcal{A}} (\mathbf{X}_{\mathcal{A}}^\top \mathbf{X}_{\mathcal{A}})^{-1} \text{sign}(\hat{\beta}(\lambda_k)_{\mathcal{A}}).$$

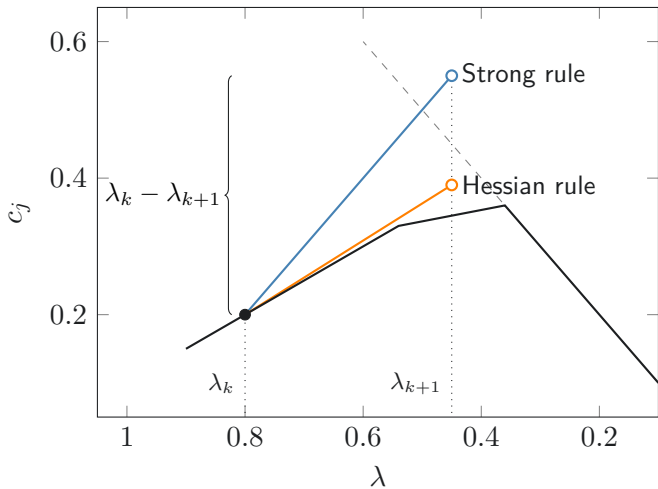


Figure 6: The strong and Hessian screening rules in action

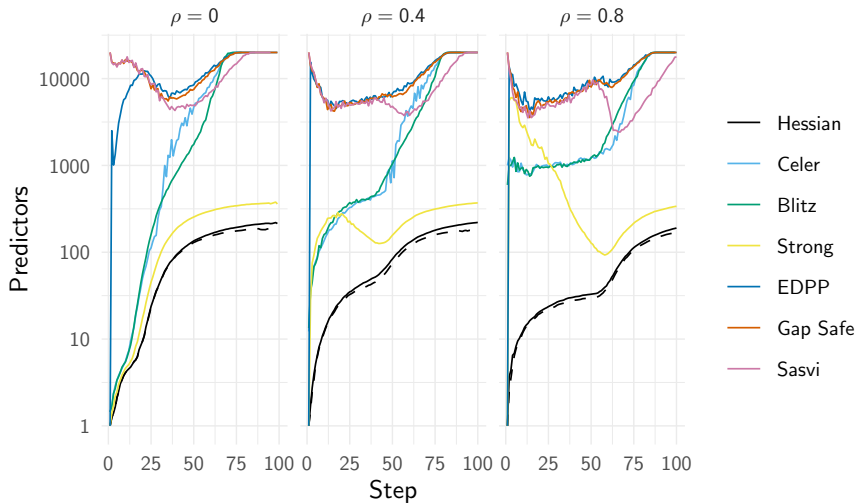


Figure 7: Features screened for a lasso path for ℓ_1 -regularized least-squares to a design with varying correlation (ρ), $n = 200$, and $p = 20\,000$.

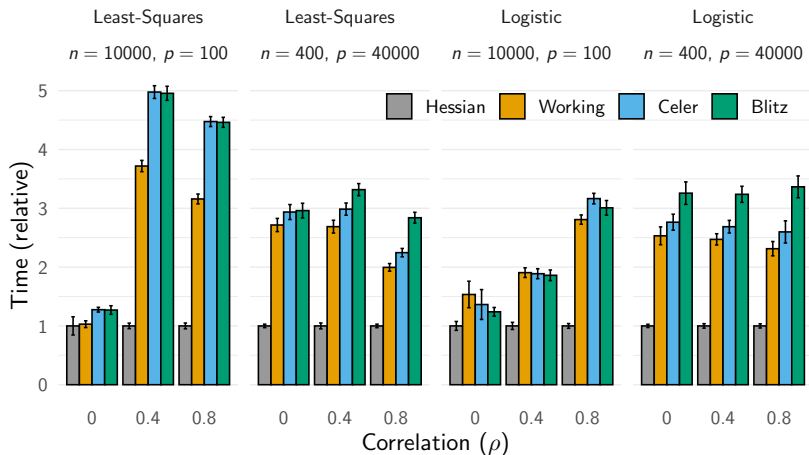


Figure 8: Time to fit a full path of ℓ_1 -regularized least-squares and logistic regression to a design with n observations, p features, and pairwise correlation between features of ρ .

The Hessian Screening Rule

Published and presented at NeurIPS
2022²

Co-authored with Jonas



²Johan Larsson and Jonas Wallin (Nov. 28–Dec. 9, 2022). “The Hessian Screening Rule”. In: *Advances in Neural Information Processing Systems 35*. 36th Conference on Neural Information Processing Systems (NeurIPS 2022). Ed. by Sanmi Koyejo et al. Vol. 35. New Orleans, USA: Curran Associates, Inc., pp. 15823–15835. ISBN: 978-1-71387-108-8

Coordinate Descent for SLOPE (Paper V)

Coordinate Descent for SLOPE

Published and presented at AISTATS 2023³



Co-authored with Mathurin Massias, Quentin Bertrand, and Jonas Wallin.

³Johan Larsson, Quentin Klopfenstein, et al. (Apr. 25–27, 2023). “Coordinate Descent for SLOPE”. In: *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*. AISTATS 2023. Ed. by Francisco Ruiz, Jennifer Dy, and Jan-Willem van de Meent. Vol. 206. Proceedings of Machine Learning Research. Valencia, Spain: PMLR, pp. 4802–4821

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- Lasso solvers are fast because they use coordinate descent
- Unfortunately cannot use basic coordinate descent for SLOPE

Coordinate Descent

Simple Method!

At each iteration, update a single coordinate (coefficient).

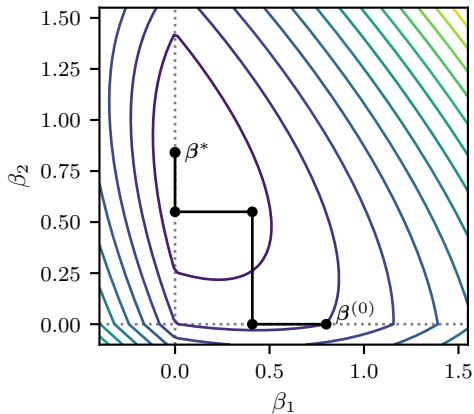


Figure 9: A basic example of coordinate descent in two dimensions

Coordinate Descent Performance

Performs (surprisingly) well!

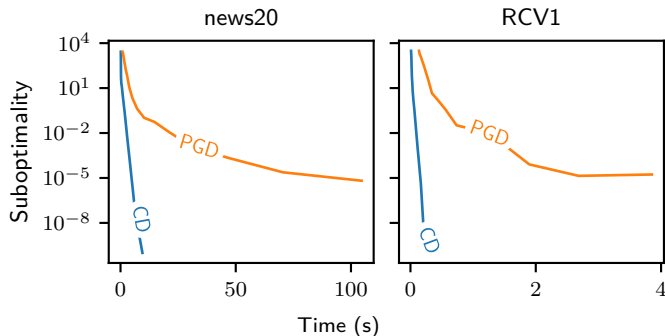


Figure 10: Coordinate descent versus proximal gradient descent for the lasso.

Coordinate Descent and Separability

Problem

Cannot use basic coordinate descent for SLOPE since the sorted ℓ_1 norm is **inseparable**:

$$h(\beta) = \sum_{j=1}^p \lambda_j |\beta_{(j)}|.$$

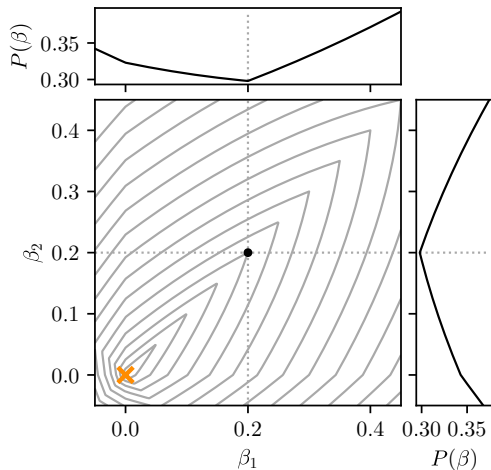


Figure 11: A *naive* coordinate descent algorithm cannot advance from the current iterate (●) to reach the optimum (×).

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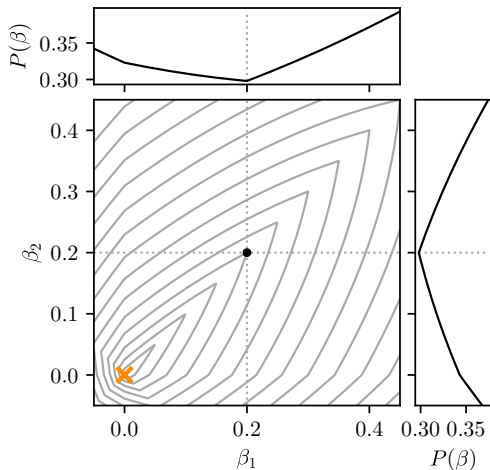


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Idea

Alternate between gradient descent and coordinate descent **on the clusters**.

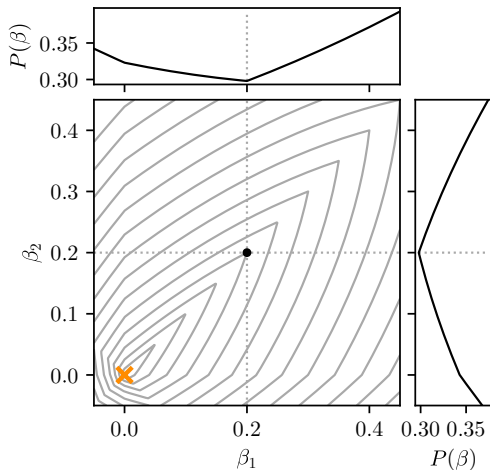


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Hybrid Algorithm

- Every v th iteration, take a full proximal gradient step. This allows clusters to split (or merge).
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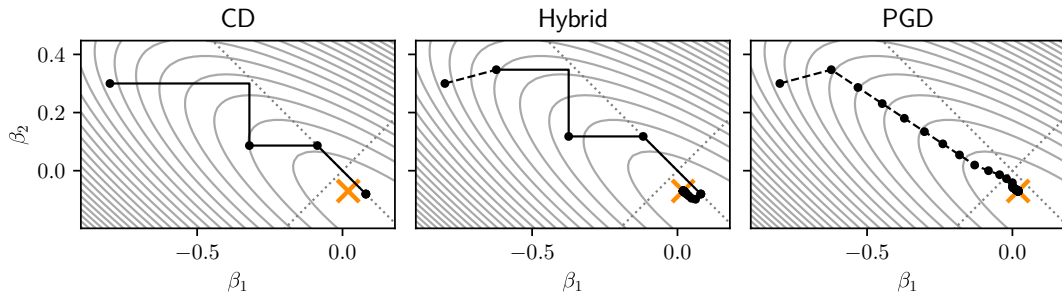


Figure 12: Our algorithm (hybrid) is a combination of CD and PGD.

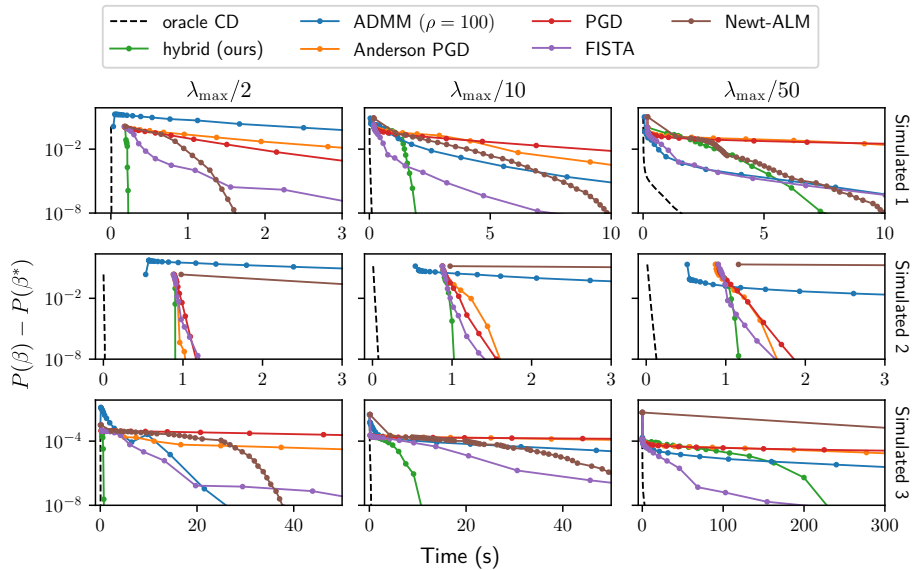
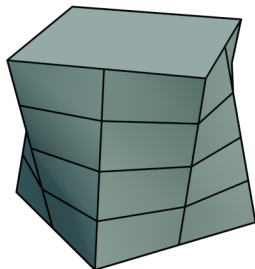


Figure 13: Benchmarks on simulated data. Scenario 1: $n = 200$ and $p = 20\,000$, X . Scenario 2: $n = 20\,000$ and $p = 200$. Scenario 3: $n = 200$, $p = 200\,000$, and sparse X .

The SLOPE Package(s)

LibSLOPE

- C++ library for solving SLOPE using the hybrid coordinate method
- Natively parallelized routines for fitting paths and cross-validation
- Algorithmic improvements to SLOPE thresholding and cluster handling for sparse data



Packages

R SLOPE: <https://jolars.github.io/slope/>

Python sortedl1: <https://jolars.github.io/sortedl1/>

C++ libslope: <https://jolars.github.io/sortedl1/>

The Choice of Normalization Affects Shrinkage in Regularized Regression

The Elastic Net

Linear regression plus a combination of the ℓ_1 and ℓ_2 penalties:

$$\boldsymbol{\beta}^* = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \underbrace{\lambda_1 \|\boldsymbol{\beta}\|_1}_{\text{lasso}} + \underbrace{\frac{\lambda_2}{2} \|\boldsymbol{\beta}\|_2^2}_{\text{ridge}} \right)$$

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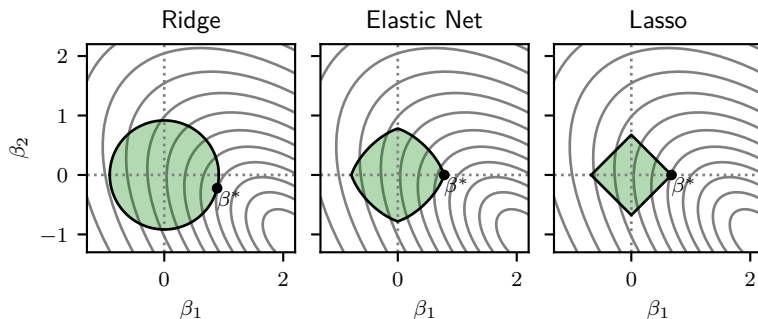


Figure 14: Elastic net is a combination of the lasso and ridge.

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Scale Sensitivity

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Example

Assume

$$\mathbf{X} \sim \text{Normal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \right), \quad \boldsymbol{\beta}^* = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}.$$

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Model	$\hat{\boldsymbol{\beta}}$	$\hat{\boldsymbol{\beta}}_{\text{std}}$
OLS	$\begin{bmatrix} 0.50 & 1.00 \end{bmatrix}^{\top}$	$\begin{bmatrix} 1.00 & 1.00 \end{bmatrix}^{\top}$

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Large scale means **less** penalization because the size of β_j can be smaller for an equivalent effect (on $\mathbf{y}$).

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- After fitting, we transform the coefficients back to their original scale via

$$\hat{\beta}_j = \frac{\hat{\beta}_j^{(n)}}{s_j} \quad \text{for } j = 1, 2, \dots, p,$$

where $\hat{\beta}_j^{(n)}$ is a coefficient from the normalized problem.

Table 2: Common ways to normalize $\mathbf{X}$

Normalization	Centering (c_j)	Scaling (s_j)
Standardization	$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$	$\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}$
Min–Max	$\min_i(x_{ij})$	$\max_i(x_{ij}) - \min_i(x_{ij})$
Unit Vector (L2)	0	$\sqrt{\sum_{i=1}^n x_{ij}^2}$
Max–Abs	0	$\max_i(x_{ij})$
Adaptive Lasso	0	β_j^{OLS}

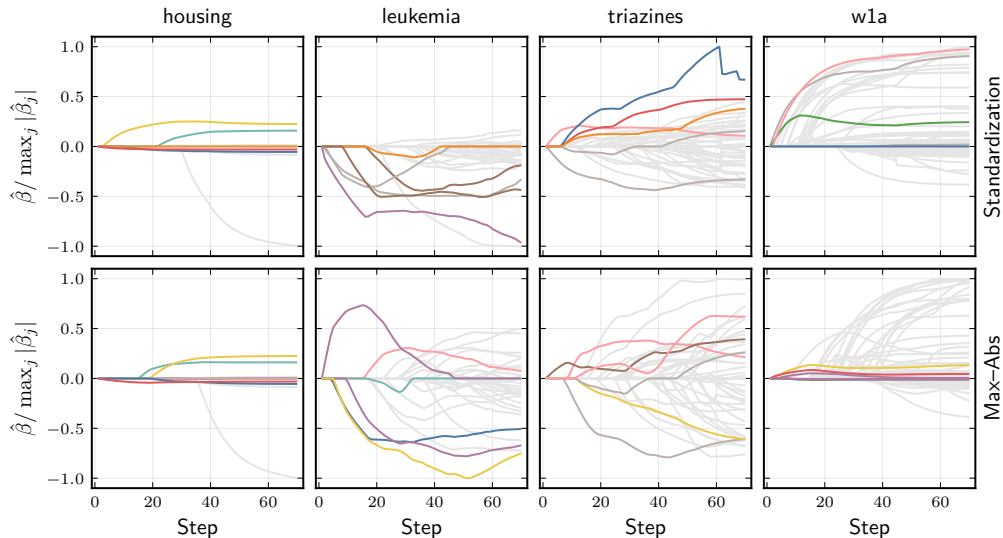


Figure 15: The motivation: normalization matters. Lasso paths under two different types of normalization (standardization and max-abs normalization). The union of the first ten features selected in any of the settings are colored.

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- Binary features, particularly with respect to the **class balance** thereof
- A mix of binary and normally distributed features

Orthogonal Features

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But if we assume that the features are orthogonal, that is

$$\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = \text{diag}(\tilde{\mathbf{x}}_1^\top \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_p^\top \tilde{\mathbf{x}}_p),$$

then there is:⁴:

$$\hat{\beta}_j = \frac{S_{\lambda_1}(\tilde{\mathbf{x}}_j^\top \mathbf{y})}{s_j(\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{x}}_j + \lambda_2)},$$

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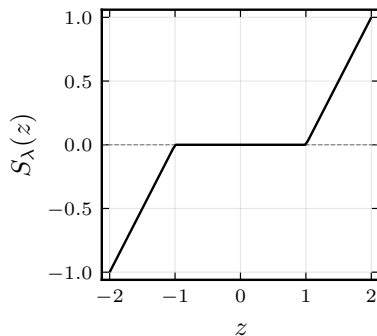


Figure 16: Soft thresholding

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Bias and Variance of the Elastic Net Estimator

The goal is computing the expected value of the elastic net estimator,

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since we treat $\mathbf{X}$ as fixed.

Letting $Z = \tilde{\mathbf{x}}^\top \mathbf{y}$ and assuming that ε_i is i.i.d. Normally-distributed with mean zero and finite variance σ_ε^2 , we have

$$Z \sim \text{Normal}(\mu = \tilde{\mathbf{x}}_j^\top \mathbf{x}_j \beta_j, \sigma^2 = \sigma_\varepsilon^2 \|\tilde{\mathbf{x}}_j\|_2^2).$$

Next, will turn to $\mathbb{E} S_{\lambda_1}(Z)$.

Noiseless Case for Binary Features

In the noiseless case, we have

$$\hat{\beta}_j = \frac{S_{\lambda_1}(\tilde{\mathbf{x}}^\top \mathbf{y})}{s_j (\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{x}}_j + \lambda_2)} = \frac{S_{\lambda_1} \left(\frac{\beta_j^* n (q - q^2)}{s_j} \right)}{s_j \left(\frac{n (q - q^2)}{s_j^2} + \lambda_2 \right)}.$$

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- Means that the elastic net estimator depends on class balance (q).

Noiseless Case for Binary Features

In the noiseless case, we have

$$\hat{\beta}_j = \frac{S_{\lambda_1}(\tilde{\mathbf{x}}^\top \mathbf{y})}{s_j (\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{x}}_j + \lambda_2)} = \frac{S_{\lambda_1} \left(\frac{\beta_j^* n (q - q^2)}{s_j} \right)}{s_j \left(\frac{n (q - q^2)}{s_j^2} + \lambda_2 \right)}.$$

- Means that the elastic net estimator depends on class balance (q).
- $s_j = q - q^2$ for lasso and $s_j = \sqrt{q - q^2}$ for ridge removes effect of q .

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$$s_j = (q - q^2)^\delta, \quad \delta \geq 0,$$

which we will rely on for the rest of the talk.

- Indicates there might be no (simple) s_j that will work for the elastic net.

Probability of Selection

Since $\mathbf{X}$ is fixed and ε is normal, we can compute the probability of selection:

$$\Pr(\hat{\beta}_j \neq 0) = \Phi\left(\frac{\mu - \lambda_1}{\sigma}\right) + \Phi\left(\frac{-\mu - \lambda_1}{\sigma}\right).$$

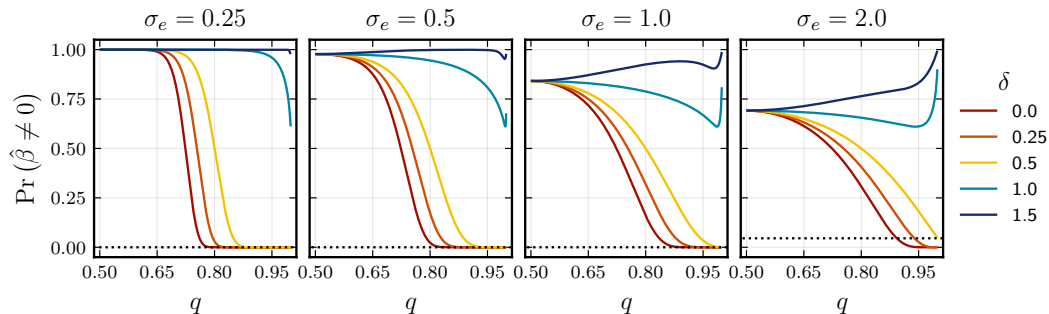


Figure 17: Probability that the elastic net selects a feature across different noise levels (σ_ε), types of normalization (δ), and class balance (q). The dashed line is asymptotic behavior for $\delta = 1/2$. Scaling used is $s_j \propto (q - q^2)^\delta$.

Rare Traits

Features with large class-imbalances might not be selected even if effect is **very strong** (e.g. rare SNPs, mutations).

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Subgroup Data

Results become dependent on data collection.

Collecting more data with different class balances influences the results (since class balances change).

Theorem

If x_j is a binary feature with class balance $q \in (0, 1)$ and $\lambda_1, \lambda_2 \in (0, \infty)$, $\sigma_\varepsilon > 0$, and $s_j = (q - q^2)^\delta$, $\delta \geq 0$, then

$$\lim_{q \rightarrow 1^+} \mathbb{E} \hat{\beta}_j = \begin{cases} 0 & \text{if } 0 \leq \delta < \frac{1}{2}, \\ \frac{2n\beta_j^*}{n+\lambda_2} \Phi\left(-\frac{\lambda_1}{\sigma_\varepsilon\sqrt{n}}\right) & \text{if } \delta = \frac{1}{2}, \\ \beta_j^* & \text{if } \delta \geq \frac{1}{2}. \end{cases}$$

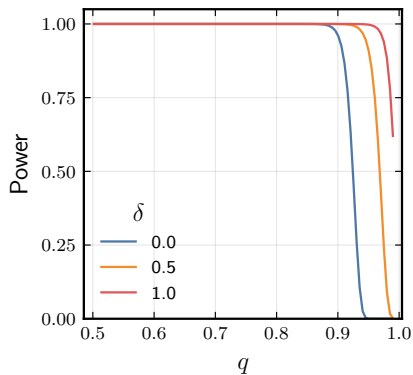
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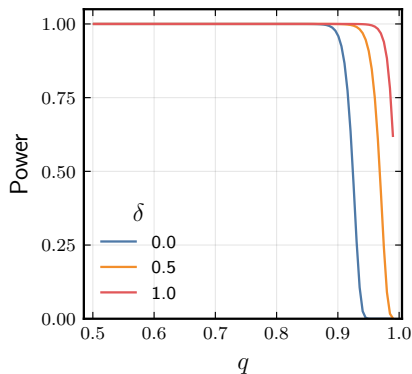
and

$$\lim_{q \rightarrow 1^+} \text{Var} \hat{\beta}_j = \begin{cases} 0 & \text{if } 0 \leq \delta < \frac{1}{2}, \\ \infty & \text{if } \delta \geq \frac{1}{2}. \end{cases}$$

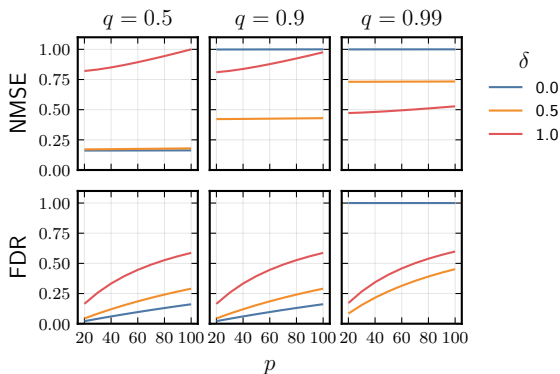


(a) Power in the sense of detecting all the true signals. Constant p .

Figure 18: Multiple features: 10 true signals and varying q and p . Mean squared error (MSE), false discovery rate (FDR), and power



(a) Power in the sense of detecting all the true signals. Constant p .



(b) False discovery rate (FDR) and normalized mean-squared error (NMSE).

Figure 18: Multiple features: 10 true signals and varying q and p . Mean squared error (MSE), false discovery rate (FDR), and power

Mixed Data

So far: all binary features. What about mixing binary and continuous (normal) features?

How to put binary features and normal features on the “same” scale?

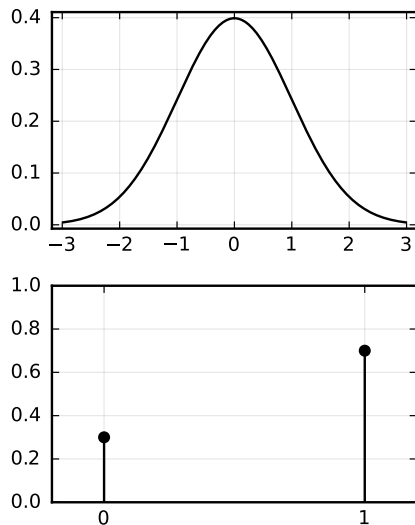


Figure 19: How do we match these?

Our Definition of Comparability

The effects of a binary feature and a normally distributed feature are **comparable** if a flip in the binary feature has the same effect as a two-standard deviation change in the normal feature (**gelman2008**).

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Examples

Assume entries in x_1 are binary and x_2 come from a random variable X_2 . The effects are comparable in the following cases:

- $X_2 \sim \text{Normal}(\mu, 1/2)$, $\beta_1^* = 1$, and $\beta_2^* = 1$.
- $X_2 \sim \text{Normal}(\mu, 2)$, $\beta_1^* = 1$, and $\beta_2^* = 0.25$.

Choice of Scaling in Mixed Data

For the two-standard deviation notion of comparability to hold, we need to modify our scaling factor s_j .

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As before, we assume that x_1 is binary and $X_2 \sim \text{Normal}(\mu, 1/2)$, $\beta_1^* = \beta_2^* = 1$ so that they have *comparable* effects. Also assume we standardize x_2 .

We want $\hat{\beta}_1 = \hat{\beta}_2$. That is,

$$\underbrace{\frac{S_{\lambda_1} \left(\frac{n(q-q^2)}{s_j} \right)}{s_1 \left(\frac{n(q-q^2)}{s_1^2} + \lambda_2 \right)}}_{\hat{\beta}_1} = \underbrace{\frac{S_{\lambda_1} \left(\frac{n}{2} \right)}{\frac{1}{2} (n + \lambda_2)}}_{\hat{\beta}_2}.$$

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The choice $s_1 = (2(q - q^2))^\delta$ works when classes are balanced ($q = 0.5$). But no clear choice for the elastic net case.

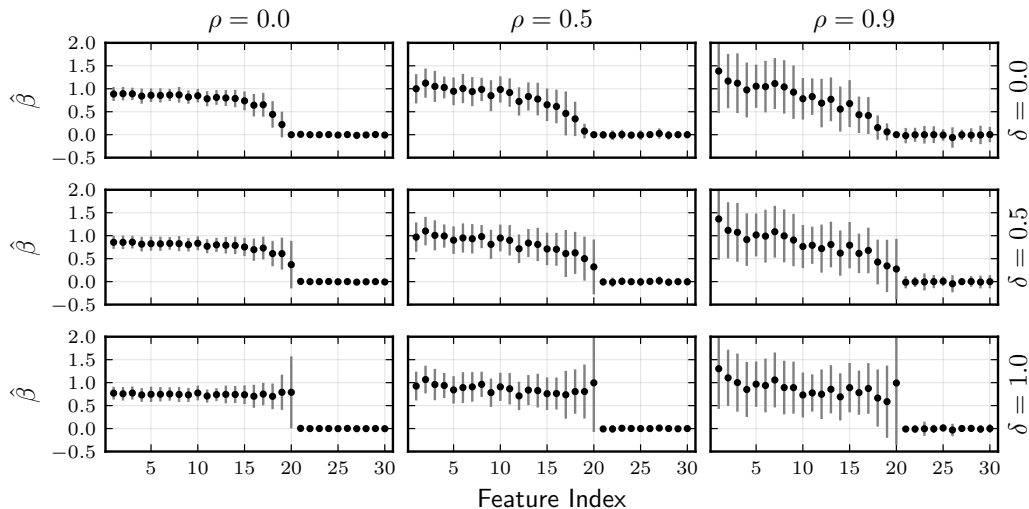


Figure 20: Lasso estimates for first 30 coefficients. First 20 features are true signals with a geometrically decreasing class balance from 0.5 to 0.99. ρ is a measure of autocorrelation.

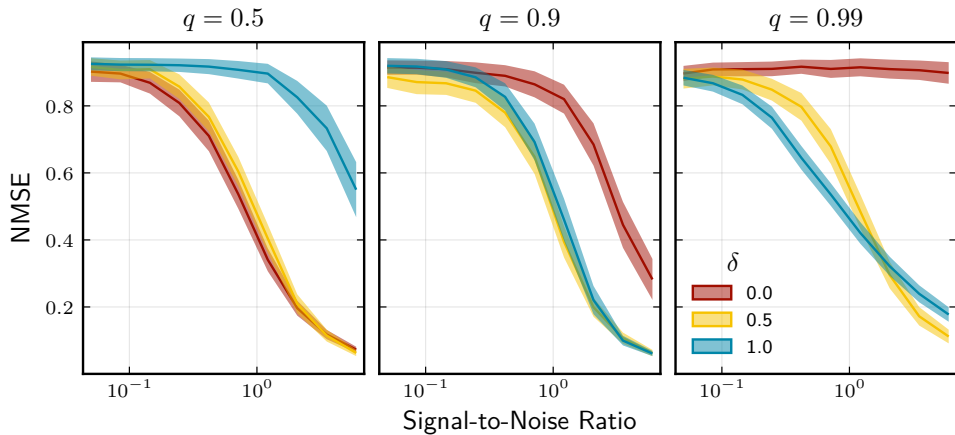


Figure 21: Normalized mean-squared test set error (NMSE).

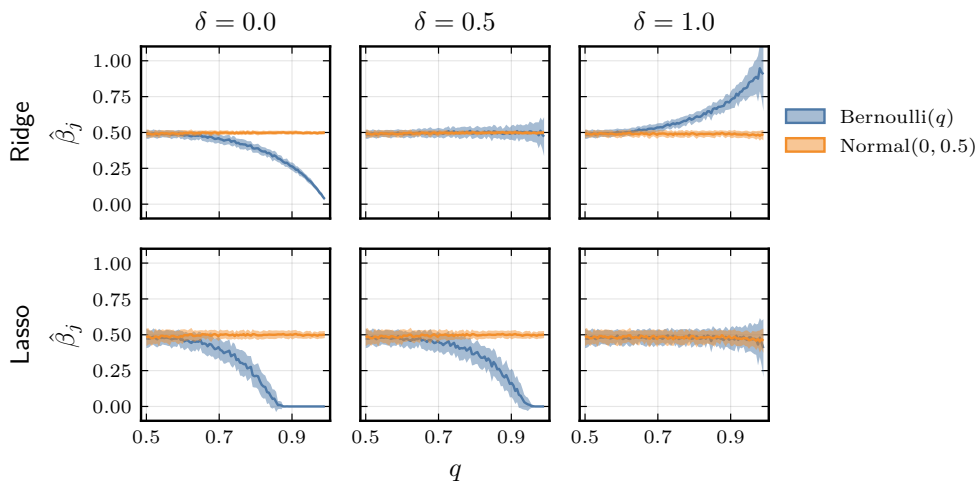


Figure 22: Comparison between lasso and ridge estimators for features generated to resemble features from various distributions.

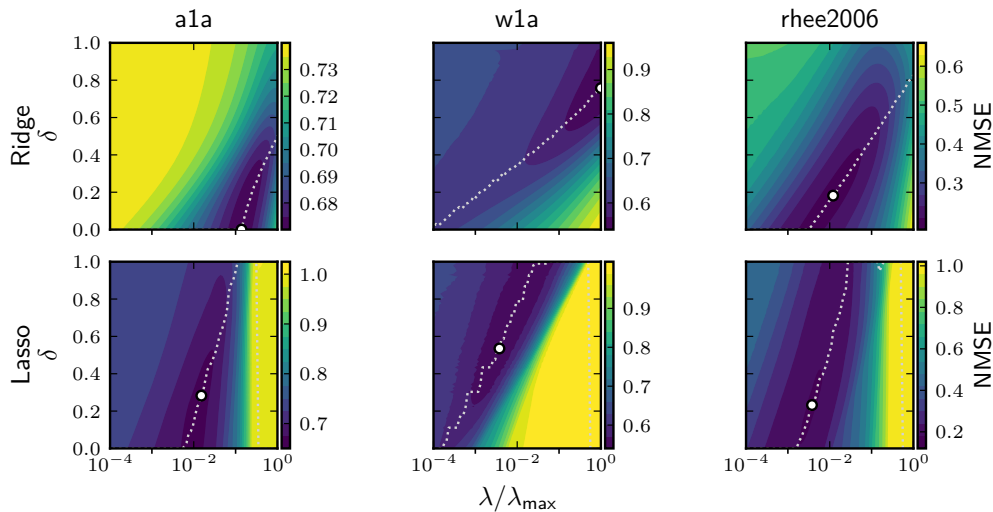


Figure 23: Contour plots of hold-out (validation set) error across a grid of δ and λ values for the lasso and ridge.

- Several screening rules that improve performance for lasso and SLOPE

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- Analysis of interplay between normalization and penalization

Thank you!

-  Larsson, Johan, Quentin Klopfenstein, et al. (Apr. 25–27, 2023). “Coordinate Descent for SLOPE”. In: *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*. AISTATS 2023. Ed. by Francisco Ruiz, Jennifer Dy, and Jan-Willem van de Meent. Vol. 206. Proceedings of Machine Learning Research. Valencia, Spain: PMLR, pp. 4802–4821.
-  Larsson, Johan and Jonas Wallin (Nov. 28–Dec. 9, 2022). “The Hessian Screening Rule”. In: *Advances in Neural Information Processing Systems 35*. 36th Conference on Neural Information Processing Systems (NeurIPS 2022). Ed. by Sanmi Koyejo et al. Vol. 35. New Orleans, USA: Curran Associates, Inc., pp. 15823–15835. ISBN: 978-1-71387-108-8.
-  Tibshirani, Robert et al. (Mar. 2012). “Strong Rules for Discarding Predictors in Lasso-Type Problems”. In: *The Journal of the Royal Statistical Society, Series B (Statistical Methodology)* 74.2, pp. 245–266. ISSN: 1369-7412. DOI: [10/c4bb85](https://doi.org/10/c4bb85).

Paper I Extras

Strong Rule Algorithm for SLOPE

Algorithm 1: Basic algorithm for the strong rule for SLOPE

Data: $\mathbf{c} \in \mathbb{R}^p$, $\boldsymbol{\lambda} \in \mathbb{R}^p$, where $\lambda_1 \geq \dots \geq \lambda_p \geq 0$

$\mathcal{S}, \mathcal{B} \leftarrow \emptyset$;

for $i \leftarrow 1$ **to** p **do**

$\mathcal{B} \leftarrow \mathcal{B} \cup \{i\}$;

if $\sum_{j \in \mathcal{B}} (c_j - \lambda_j) \geq 0$ **then**

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end

return $\mathcal{S}$;

Set

$$\mathbf{c} = |\nabla g(\hat{\boldsymbol{\beta}}) + \boldsymbol{\lambda}^{(k-1)} - \boldsymbol{\lambda}^{(k)}|_{\downarrow} \quad \boldsymbol{\lambda} = \boldsymbol{\lambda}^{(k)},$$

and run the algorithm above; the result is the predicted support for $\hat{\boldsymbol{\beta}}(\boldsymbol{\lambda}^{(k)})$ (subject to a permutation).

Results: Speed

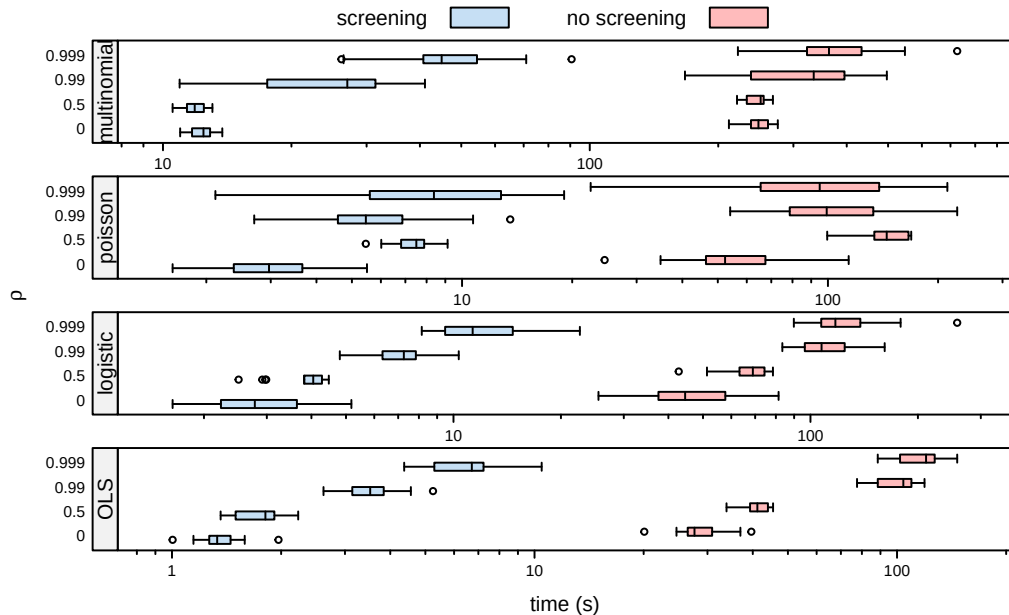


Figure 24: Time to fit a full SLOPE path with and without the strong rule

Paper II Extras

The Dual

Complementary to the primal problem. For the lasso, it is

$$\underset{\boldsymbol{\theta} \in \mathbb{R}^n}{\text{maximize}} \left(D(\boldsymbol{\theta}; \lambda) = \frac{1}{2} \mathbf{y}^\top \mathbf{y} - \frac{\lambda^2}{2} \left\| \boldsymbol{\theta} - \frac{\mathbf{y}}{\lambda} \right\|_2^2 \right)$$

The Dual

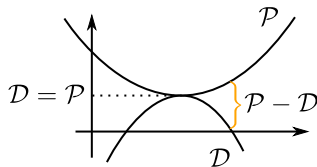
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Duality Gap

Difference between the primal and dual objectives. Tight at the optimum (for the lasso):

$$P(\boldsymbol{\beta}^*; \lambda) - D(\boldsymbol{\theta}^*; \lambda) = 0.$$



The Dual

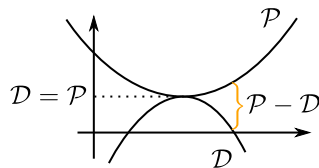
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Duality Gap

Difference between the primal and dual objectives. Tight at the optimum (for the lasso):

$$P(\boldsymbol{\beta}^*; \lambda) - D(\boldsymbol{\theta}^*; \lambda) = 0.$$



Dual-Primal Relationship

The two problems are related via

$$\frac{\mathbf{y} - \mathbf{X}\boldsymbol{\beta}}{\lambda} = \boldsymbol{\theta}.$$

Inequality for Gaps Safe screening rule is **quadratic** in λ , which means we can find the next critical point easily:

$$\lambda_+ = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where

$$a = (1 - |\mathbf{x}_j^\top \boldsymbol{\theta}_\lambda|)^2 - \frac{1}{2} \boldsymbol{\theta}_\lambda^\top \boldsymbol{\theta}_\lambda \|\mathbf{x}_j\|_2^2,$$

$$b = (\boldsymbol{\theta}_\lambda^\top \mathbf{y} - \|\boldsymbol{\beta}_\lambda\|_1) \|\mathbf{x}_j\|_2^2,$$

$$c = -\frac{1}{2} \|\mathbf{y} - \mathbf{X} \boldsymbol{\beta}_\lambda\|_2^2 \|\mathbf{x}_j\|_2^2.$$

Look-Ahead Screening

Inequality for Gaps Safe screening rule is **quadratic** in λ , which means we can find the next critical point easily:

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$$c = -\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_\lambda\|_2^2 \|\mathbf{x}_j\|_2^2.$$

Allows us to screen predictors for **all** upcoming steps

Paper III Extras

The availability of the Hessian inverse enables a better warm start:

$$\hat{\beta}(\lambda_{k+1})_{\mathcal{A}} = \hat{\beta}(\lambda_k)_{\mathcal{A}} + (\lambda_k - \lambda_{k+1})(X_{\mathcal{A}}^T X_{\mathcal{A}})^{-1} \text{sign}(\hat{\beta}(\lambda_k)_{\mathcal{A}})$$

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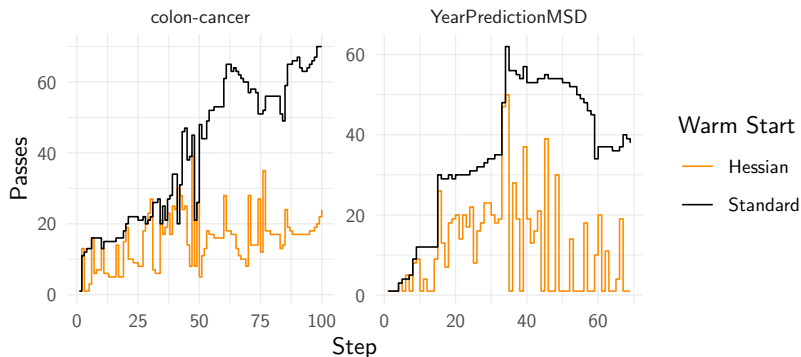


Figure 25: Number of passes of coordinate descent for two datasets using either Hessian warm starts or standard warm starts.

Computing the Hessian and its inverse naively is expensive: $\mathcal{O}(|\mathcal{A}|^3 + |\mathcal{A}|^2 n)$

Updating the Hessian

Computing the Hessian and its inverse naively is expensive: $\mathcal{O}(|\mathcal{A}|^3 + |\mathcal{A}|^2 n)$

Fortunately, we can sweep columns of the Hessian and inverse in our out, yielding complexity

- $\mathcal{O}(|\mathcal{D}|^2 n + n|\mathcal{D}||\mathcal{E}| + |\mathcal{D}|^2|\mathcal{E}| + |\mathcal{D}|^3)$ when augmenting the Hessian and
- $\mathcal{O}(|\mathcal{C}|^3 + |\mathcal{C}|^2|\mathcal{E}| + |\mathcal{C}||\mathcal{E}|^2)$ when reducing it,

where

- $\mathcal{C} = \mathcal{A}_{k-1} \setminus \mathcal{A}_k$ (to-be deactivated)
- $\mathcal{D} = \mathcal{A}_k \setminus \mathcal{A}_{k-1}$ (to-be activated)
- $\mathcal{E} = \mathcal{A}_k \cap \mathcal{A}_{k-1}$ (still activate)

Paper V Extras

How Do We Minimize Over One Cluster?

The optimality condition, using the directional derivative, is

$$\forall \delta \in \{-1, 1\}, \quad G'(z; \delta) \geq 0,$$

with

$$\begin{aligned} G'(z; \delta) &= \delta \sum_{j \in \mathcal{C}_k} X_{:,j}^\top (X\beta(z) - y) \\ &\quad + H'(z; \delta). \end{aligned}$$

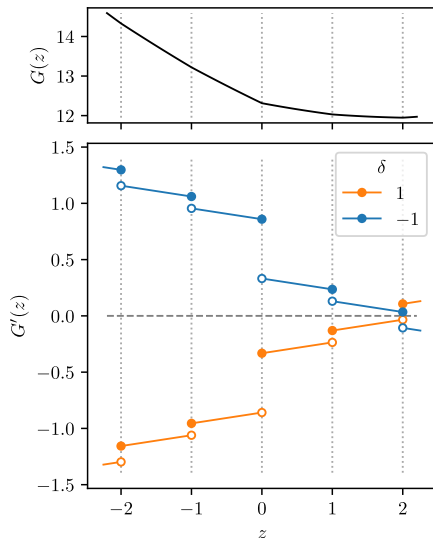


Figure 26: G and its directional derivative $G'(\cdot; \delta)$ for an example with $\beta = [-3, 1, 3, 2]^T$, $k = 1$, and consequently

The SLOPE Thresholding Operator

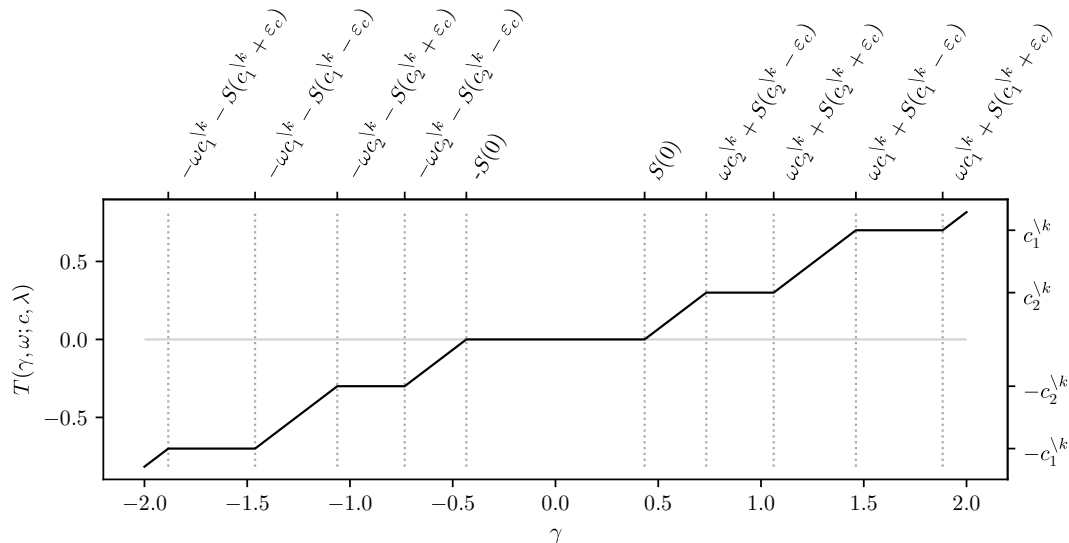


Figure 27: The SLOPE Thresholding Operator

Experiments: Real Data

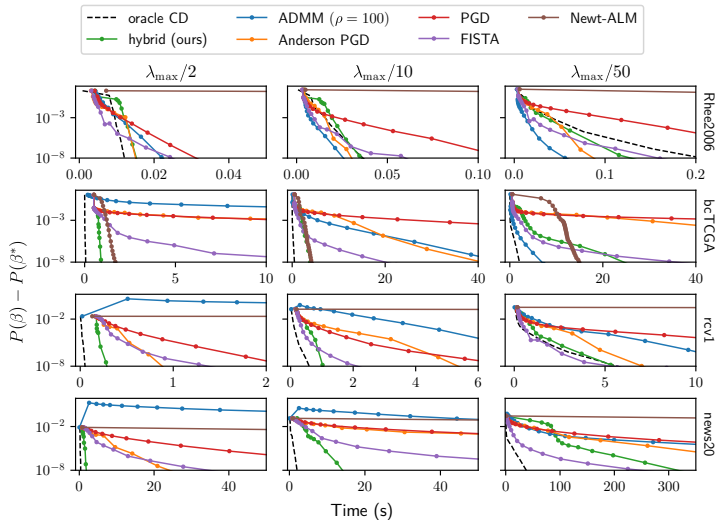


Figure 28: Benchmarks on real data

Paper VI Extras

Probability of Selection

Since $\mathbf{X}$ is fixed and ε is normal, it is straightforward to compute the probability of selection:

$$\Pr(\hat{\beta}_j \neq 0) = \Phi\left(\frac{\mu - \lambda_1}{\sigma}\right) + \Phi\left(\frac{-\mu - \lambda_1}{\sigma}\right).$$

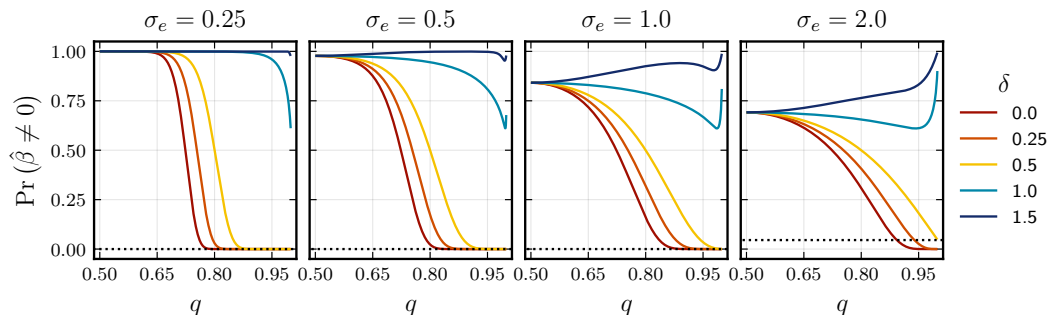


Figure 29: Probability that the elastic net selects a feature across different noise levels (σ_ε), types of normalization (δ), and class balance (q). The dashed line is asymptotic behavior for $\delta = 1/2$.

Theorem

If x_j is a binary feature with class balance $q \in (0, 1)$ and $\lambda_1, \lambda_2 \in (0, \infty)$, $\sigma_\varepsilon > 0$, and $s_j = (q - q^2)^\delta$, $\delta \geq 0$, then

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Theorem

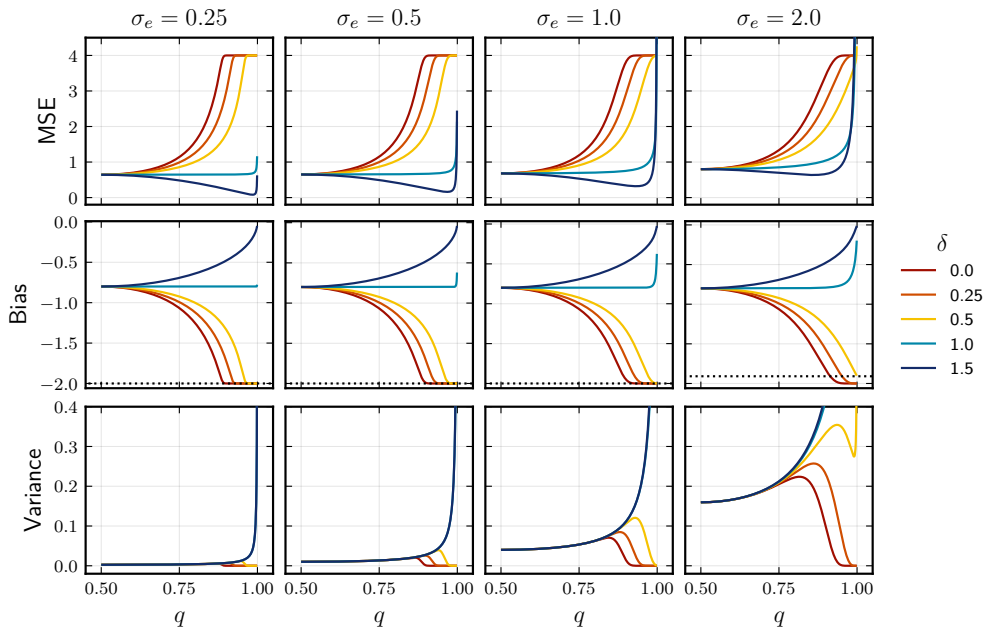
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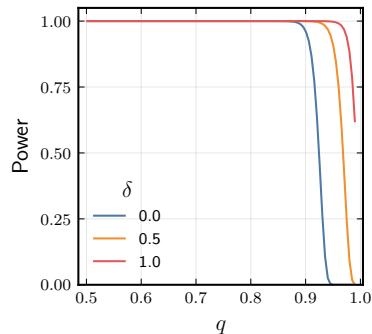
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A Bias–Variance Tradeoff



Multiple Features: Power, FDR, and NMSE

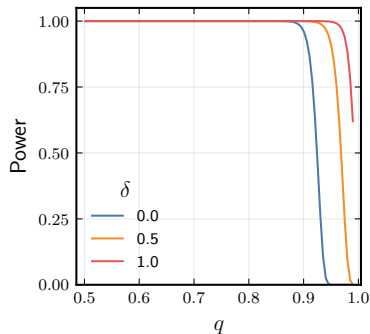
Lasso example with 10 true signals and varying q and p .



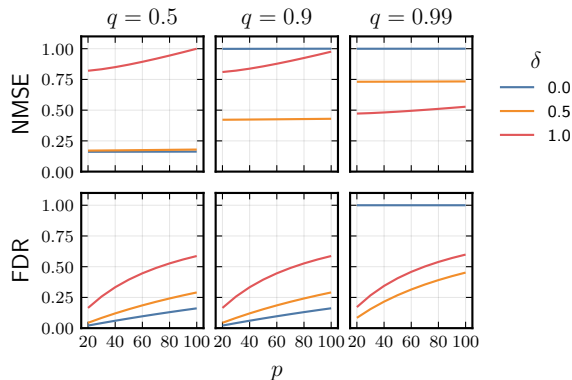
(a) Power in the sense of detecting all the true signals. Constant p .

Multiple Features: Power, FDR, and NMSE

Lasso example with 10 true signals and varying q and p .



(a) Power in the sense of detecting all the true signals. Constant p .



(b) False discovery rate (FDR) and normalized mean-squared error (NMSE).

Figure 31: Mean squared error (MSE), false discovery rate (FDR), and power.

Hyperparameter Optimization

Idea: The choice of δ affects the model, so let's optimize over it.

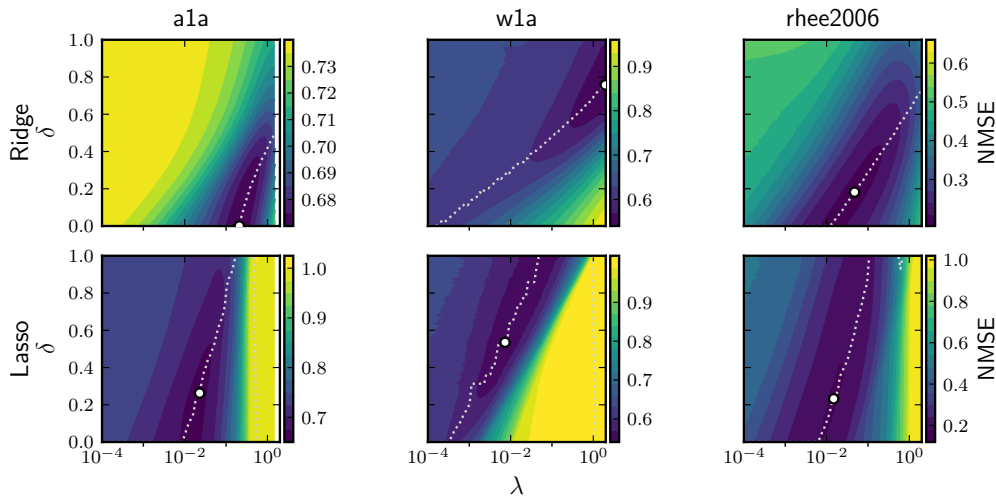


Figure 32: Contour plots of hold-out (validation set) error across a grid of δ and λ values for the lasso and ridge.